

Algebra Practice

Synopsis: Math is an integral part of doing physics at any level. We handle equations or algebraic expressions. You are accustomed to doing algebra in your head, although perhaps you don't know it.

Imagine you are in a car driving on a highway at 60 mph. Your destination is 120 miles away. How many hours will it take to get there? You immediately say, 2 hours. To get this answer, you did an algebraic manipulation.

To see how, remember that speed is the distance covered divided by the time taken.

$$\text{Speed (miles/hour)} = \frac{\text{Distance (miles)}}{\text{Time (hour)}}, \text{ that is, } 60 \text{ (miles/hour)} = \frac{60 \text{ (miles)}}{1 \text{ (hour)}}$$

To calculate the time taken, you took the total distance of 120 miles and divided it by the speed, 60 miles per hour.

$$\text{Time (hours)} = \frac{\text{Distance (miles)}}{\text{Speed (miles/hour)}}, \text{ that is, } \text{Time (hours)} = \frac{120 \text{ (miles)}}{60 \text{ (miles/hour)}} = 2 \text{ hours}$$

Let's use symbols, Speed = s , Distance = d , and Time = t , and rewrite the above expressions:

$$s = \frac{d}{t} \quad \text{and} \quad t = \frac{d}{s}$$

I want you to notice what changed from the first to the second expression: s in the numerator (“upstairs”) became the denominator (“downstairs”), and t in the denominator (“downstairs”) became the numerator (“upstairs”) when s and t “crossed” over to the other side of the equal sign.

$$\begin{aligned} \frac{s}{1} &= \frac{d}{t} \\ \frac{t}{1} &= \frac{d}{s} \end{aligned}$$

Note that any number divided by 1 is itself, hence $s = s/1$.

So, this particular manipulation involved two simultaneous “crossing overs.” But it doesn't have to be. Let's just cross the t over and leave s .

$$\begin{aligned} \frac{s}{1} &= \frac{d}{t} \\ \frac{st}{1} &= \frac{d}{1} \end{aligned}$$

Or $d = st$. In other words, you just derived and expressed the distance covered, d , when you travel at speed, s , for time, t . Using the example above, $(60 \text{ miles/hour}) \times (2 \text{ hours}) = 120 \text{ miles}$.

This “trick,” when you cross over the equal sign, you go from downstairs to upstairs and vice versa, will always work because it shortcuts the many steps of multiplying or dividing both sides that you are used to doing. You should convince yourself that you would get the same answer if you just did it the way you were taught in middle school. Don't bother with middle school procedures once you are convinced and comfortable with this method.

There is a similar shortcut involving addition and subtraction. Again, you are used to doing the algebra in your head. Imagine you are in a class of 25. There are 10 males. How many females are there? You immediately say 15. What algebraic manipulation did you do?

Let total = t , males = m , and females = f , so $t = m + f$. That is $25 = 10 + f$. To find f , you did $25 - 10 = 15$, the female number. Or algebraically, $f = t - m$.

The similar trick I am referring to is this: If you take a term (t , f , and m are the terms) and “throw it over” to the other side of the equal sign, it flips the sign.

$$\begin{aligned} (+t) &= (+m) + (+f) \\ (+t) - m &= (+f) \end{aligned}$$

Note that a number without an explicit negative sign in front is a positive number. Let's do something weird. Let's throw the t over to the other side:

$$\begin{aligned} (+t) &= (+m) + (+f) \\ 0 &= (+m) + f - t \end{aligned}$$

Note that you must write a zero on the left-hand side when there are no terms or numbers! What does this look like with numbers?

$$\begin{aligned} 0 &= (+m) + f - t \\ 0 &= 10 + 15 - 25 \end{aligned}$$

You see that $0 = m + f - t$ makes total sense. The key message is this. If you throw a term over, positive becomes negative and vice versa.

Let's try something that has both addition/subtraction and multiplication/division with multiple terms.

$$d = d_0 + vt$$

Let's solve for v (or isolate v or make v the subject of the equation). There are two terms (d_0 and vt) on the right-hand side and one term (d) on the left-hand side (remember the terms are separated by + or – signs). If the symbols are multiplied or divided, they are grouped as being in a single term. Hence, vt is a single term consisting of v and t being multiplied.

Here is the sequence of operations you ought to follow:

1) Isolate the TERM that contains the symbol of interest. That is, “throw over” all terms you are not interested in solving. Here, you need to isolate the vt term, so throw d_0 over and switch the sign of d_0 .

$$\begin{aligned} d &= d_0 + vt \\ -d_0 + d &= +vt \end{aligned}$$

2) Divide out the FACTOR you do not want. Here that factor is t . BUT BEFORE YOU DIVIDE OUT, BRACKET TOGETHER ALL TERMS ON THE OPPOSITE SIDE.

$$\frac{(-d_0 + d)}{1} = \frac{+vt}{1}$$

$$\frac{(-d_0 + d)}{t} = \frac{+v}{1}$$
$$v = \frac{d - d_0}{t}$$

The last line is the same expression written more neatly.

Solve the following equations. These are equations we will actually use during the semester. Don't worry about the meanings of the variables yet — we will look at the details as the semester progresses. For now, just solve each one to the best of your ability, and ask for help on the ones you don't remember how to do.

1) Solve for f .

$$v = f \cdot \lambda$$

2) Solve for T .

$$f = \frac{1}{T}$$

3) Solve for f_s .

$$\frac{f_o}{V - v_o} = \frac{f_s}{V - v_s}$$

4) Solve for v_s .

$$\frac{f_o}{V - v_o} = \frac{f_s}{V - v_s}$$

5) Solve for t .

$$\mathbf{V} = \mathbf{V}_o + 0.610 \mathbf{t}$$

6) Solve for t .

$$\mathbf{P} = \frac{\mathbf{E}}{t}$$

7) Solve for E .

$$\mathbf{P} = \frac{\mathbf{E}}{t}$$

8) Solve for I_s .

$$\mathbf{I} = \frac{\mathbf{I}_s}{4\pi \mathbf{R}^2}$$

9) Solve for R .

$$\mathbf{I} = \frac{\mathbf{I}_s}{4\pi\mathbf{R}^2}$$

10) Solve for V_o .

$$\mathbf{V} = \mathbf{V}_o + 0.610\mathbf{t}$$